

Day 1 Problem Set

Methods Camp | University of Texas at Austin

Day 1 Problem Set

This problem set contains exercises from Day 1 that were originally done in small groups. To reinforce your understanding, please complete these exercises independently.

Exercise 1: Functions and Operations

- 1.1 Let $f(x) = 5x - 4$. Compute $f(2)$ and $f(0)$.
- 1.2 Let $f(x, y) = 3x + y^2$. Compute $f(2, 3)$.
- 1.3 Simplify: $\ln(50) - \ln(10)$. (Hint: use a log rule.)
- 1.4 Solve for x : $\ln(x) = 2$. (Hint: rewrite in exponential form.)
- 1.5 A class survey of 6 students records number of siblings: $\{1, 0, 2, 3, 1, 2\}$.
 - a. Write this as a summation expression and compute $\sum_{i=1}^6 x_i$.
 - b. Compute the mean $\bar{x}$.

Exercise 2: Ratios, Proportions, Rates & Percentages

A town has 5,400 women and 4,600 men.

- 2.1 Compute the sex ratio (males per 100 females).
- 2.2 Compute the proportion of the town that is female.
- 2.3 Last year the town recorded 27 burglaries. Compute the burglary rate per 100,000 residents. (Total population = 10,000.)
- 2.4 The town's unemployment rate fell from 8% to 6%.

- a. What is the change in percentage points?
- b. What is the percent (relative) change?

2.5 A local news headline says: “Unemployment drops 2%!” Is this headline accurate? Why or why not?

Exercise 3: Logarithms and Income

A researcher studies how income relates to years of education (X), using a model of log income:

$$\ln(Y) = 9.00 + 0.08X$$

where Y is annual income in dollars.

- 3.1 For a respondent with $X = 12$ years of education, compute $\ln(Y)$.
- 3.2 Convert $\ln(Y)$ into Y (dollars) by exponentiating. (Hint: use the inverse of $\ln$.)
- 3.3 Now consider a respondent with $X = 13$. Compute their $\ln(Y)$ and their Y .
- 3.4 Exponentiate the coefficient on X : $e^{0.08} \approx 1.083$. Interpret this number in one sentence.